

Tutorial 1 Quantities and Measurement Suggested Solutions

D1 Base units of b = base units of $V = \text{m}^3$

$$\text{Base units of } P = \frac{\text{base unit of } a}{\text{base unit of } V^2}$$

$$\text{Base units of } a = (\text{base units of } P) (\text{base units of } V^2) = (\text{kg m s}^{-2} \cdot \text{m}^{-2}) \cdot \text{m}^6 = \mathbf{\text{kg m}^5 \text{s}^{-2}}$$

$$\text{Base units of } c = (\text{base units of } P) (\text{base units of } V) = \text{kg m}^{-1} \text{s}^{-2} \cdot \text{m}^3 = \mathbf{\text{kg m}^2 \text{s}^{-2}}$$

D2 (a) Base units of $\rho = \frac{\text{unit of mass}}{\text{unit of volume}} = \mathbf{\text{kg m}^{-3}}$

(b) Base units of pressure $P = \frac{\text{unit of force}}{\text{unit of area}} = (\text{kg m s}^{-2}) \cdot \text{m}^{-2} = \mathbf{\text{kg m}^{-1} \text{s}^{-2}}$ (shown)

$$\text{(c) Base unit of } c = \left(\frac{\text{kg m}^{-1} \text{s}^{-2}}{\text{kg m}^{-3}} \right)^{\frac{1}{2}} = (\text{m}^{3-1} \text{s}^{-2})^{\frac{1}{2}} = \mathbf{\text{m s}^{-1}}$$

(d) c could be the **speed** of gas particles.

D3 (a) Base units of LHS = m^1

$$\text{Base units of RHS} = (1) (\text{m s}^{-2})^x (\text{s}^y) = \text{m}^x \text{s}^{y-2x}$$

Comparing indices, **$x = 1$ and $y = 2$** .

(b) $\therefore \mathbf{s = k a t^2}$

(c) Unit analysis will not be able to give the value of k . It can only be determined experimentally or through a more rigorous derivation.

D4 (a) $756 + 37.2 + 0.83 + 2.5 = 796.5 = \mathbf{797 \text{ (no d.p.)}}$

(largest uncertainty is ± 1 from the number 756, which is larger than the other uncertainties of ± 0.1 , ± 0.01 , ± 0.1 respectively)

Note:

When adding or subtracting data, the d.p. of the result follows that of the data with the least d.p. Hence, it is not meaningful to use different instruments to measure the length of different portions of an object. The lower number of d.p. for those parts measure with instruments of lower resolution render the high number of d.p. for other parts meaningless.

(b) 1.637 (3 d.p.)

Follow the data with the least d.p. Hence, final answer is quoted to **3 d.p.**

(c) 11 (following least sf rule) or 11.4 (using error analysis)

(i) Following least sf rule: Ans = $3.2 \times 3.563 = 11.4016 = \underline{11}$ (2 s.f.)

OR

(ii) Using error analysis, assuming absolute uncertainty is the resolution of the instrument,

$$\Delta \text{Ans} = \left(\frac{0.1}{3.2} + \frac{0.001}{3.563} \right) \times 11.4016 = 0.4$$

$$\text{Hence, Ans} = (11.4 \pm 0.4)$$

(d) 18 (following least s.f.) or 17.6 (using error analysis)

Ans can be quoted to 2 s.f. or 3 s.f., depending on whether the student uses error analysis or rough estimate by following least sf (5.6 is 2 s.f., but π has many s.f.).

Note: Rule of thumb for quick gauge

For addition /subtraction, answer follows d.p. of data with the least d.p.

For multiplication / division, answer follows the s.f. of data with least s.f.

D5

(a) Estimates:

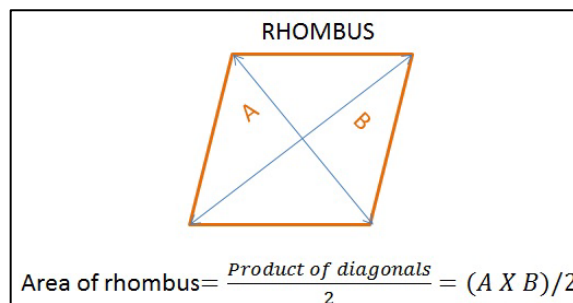
E-W breadth $b \approx 50$ km ;

N-S width $w \approx 30$ km ;

shape of the island is a rhombus

land area

$$= \frac{(50000)(30000)}{2} = 7.5 \times 10^8 \text{ m}^2$$



(b) Estimates:

Initial speed of MRT train $u \approx 0$; Final speed $v \approx 80 \text{ km h}^{-1}$; time taken $t \approx 20 \text{ s}$

$$\text{acceleration } a = \frac{v - u}{t} = \frac{80000}{60 \times 60} \times \frac{1}{20} = 1.1 \text{ m s}^{-2}$$

(c) Estimates:

Mass of car $m \approx 1000 \text{ kg}$

Car accelerates from $u \approx 60 \text{ km h}^{-1}$ to $v \approx 90 \text{ km h}^{-1}$ in $t \approx 3 \text{ s}$

$$\text{Gain in } E_k = \frac{1}{2} m (v^2 - u^2) = \frac{1}{2} (1000) \left[\left(\frac{90000}{60 \times 60} \right)^2 - \left(\frac{60000}{60 \times 60} \right)^2 \right] = 1.7361 \times 10^5 \text{ J}$$

$$\text{Power } P = \frac{\Delta E_k}{\Delta t} = \frac{1.7361 \times 10^5}{3} = 5.7870 \times 10^4 \text{ W} \approx 6 \times 10^4 \text{ W}$$

D6 **Note:** Always re-arrange the equation such that only the subject of interest is on the LHS.

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$g = \frac{4\pi^2 L}{T^2} = \frac{4\pi^2 \times 0.990}{2.00^2} = 9.771 \text{ m s}^{-2}$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T} = \frac{0.001}{0.990} + 2 \left(\frac{0.01}{2.00} \right) = 0.0110$$

$$\Delta g = 0.0110 \times 9.771 = 0.1 \text{ m s}^{-2} \text{ (to 1 s.f.)}$$

$$\therefore g = (9.8 \pm 0.1) \text{ m s}^{-2} \text{ (both values to the same 1 d.p.)}$$

D7 (a) (i) $s = ut + \frac{1}{2}at^2$

$$2.66 = 0 + \frac{1}{2}g(0.740)^2 \Rightarrow g = 9.72 \text{ m s}^{-2} \text{ (3sf)}$$

(ii) 1. $\frac{\Delta h}{h} = \frac{1}{266} \times 100\% = 0.38\% \text{ (2sf)}$

2. $\frac{\Delta t}{t} = \frac{0.005}{0.740} \times 100\% = 0.68\% \text{ (2sf)}$

(b) $s = \frac{1}{2}gt^2$

$$g = \frac{2s}{t^2} = \frac{2h}{t^2} = 9.72 \text{ m s}^{-2} \text{ (a)(i)}$$

$$\frac{\Delta g}{g} = \frac{\Delta h}{h} + 2 \frac{\Delta t}{t} = 0.38\% + 2(0.68\%) = 1.74\%$$

$$\Delta g = \frac{1.74}{100} \times g = 0.2 \text{ m s}^{-2} \text{ (1 s.f.)}$$

$$\therefore g = (9.7 \pm 0.2) \text{ m s}^{-2} \text{ (1 d.p.)}$$

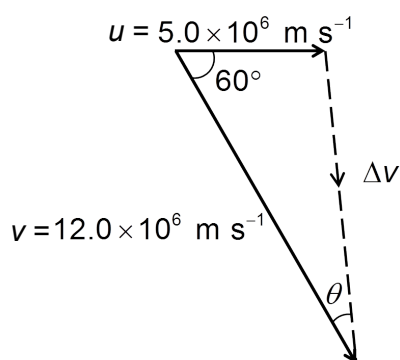
(c) For a measurement to be precise but not accurate, suggests that systematic error is involved. Possible causes for systematic error in timing may be as follows:

- A consistent time delay in anticipating the stop positions of the ball.
- Zero error in the stopwatch, i.e., the stopwatch may have been wrongly calibrated.

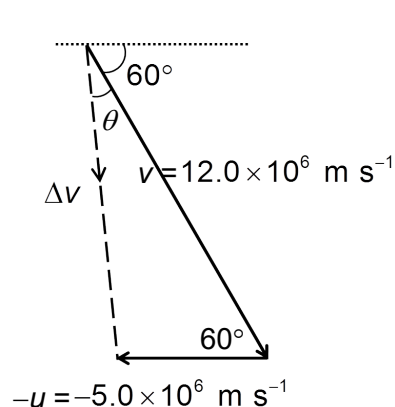
D8 Taking direction away from the racquet as positive,
 $\Delta v = v_{\text{final}} - v_{\text{initial}} = 35 - (-25) = 60 \text{ m s}^{-1}$, away from the racquet.

D9

There are two ways to calculate this via vector addition:



$$u + \Delta v = v$$



$$v - u = \Delta v$$

(a) Applying cosine rule,

$$(\Delta v)^2 = (5.0 \times 10^6)^2 + (12.0 \times 10^6)^2 - 2(5.0 \times 10^6)(12.0 \times 10^6) \cos 60^\circ$$

$$\Delta v = 10.4 \times 10^6 \text{ m s}^{-1}$$

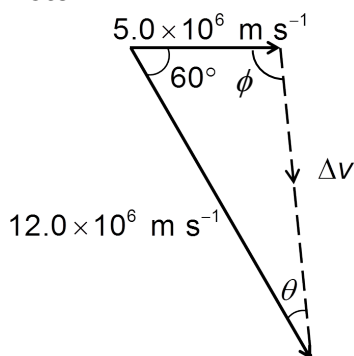
(b)
$$a = \frac{\Delta v}{\Delta t} = \frac{10.4 \times 10^6}{2.0 \times 10^{-7}} = 5.22 \times 10^{13} \text{ m s}^{-2}$$

Applying sine rule,

$$\frac{\sin \theta}{5.0 \times 10^6} = \frac{\sin 60^\circ}{10.4 \times 10^6} \Rightarrow \theta = 24.5^\circ$$

Since acceleration takes place in the direction of Δv , the acceleration of the electron is $5.22 \times 10^{13} \text{ m s}^{-2}$ at 84.5° clockwise from its initial velocity.

Note:



$$\frac{\sin \phi}{12.0 \times 10^6} = \frac{\sin 60^\circ}{10.44 \times 10^6} \Rightarrow \phi = 84.5^\circ \text{ (rejected) or } 95.5^\circ$$

The student needs to recognize that the horizontal component of the final velocity
 $= 12.0 \times 10^6 \cos 60^\circ$
 $= 6.0 \times 10^6 \text{ m s}^{-1}$
 is larger than the initial velocity.

Hence, the angle ϕ is an obtuse angle.

D10 Take right and upwards as positive.

Horizontally

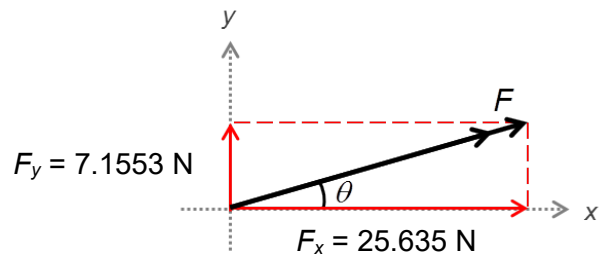
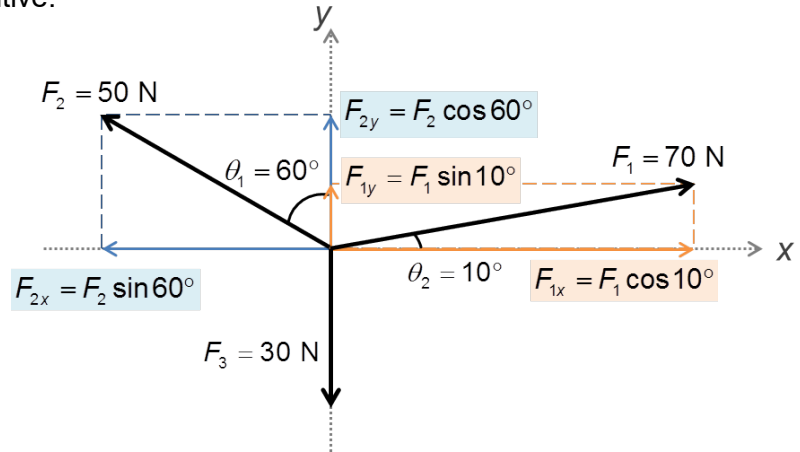
$$\begin{aligned} F_x &= F_{1x} - F_{2x} \\ &= F_1 \cos 10^\circ - F_2 \sin 60^\circ \\ &= 70 \cos 10^\circ - 50 \sin 60^\circ \\ &= 25.635 \text{ N} \end{aligned}$$

Vertically

$$\begin{aligned} F_y &= F_{1y} + F_{2y} - F_3 \\ &= F_1 \sin 10^\circ + F_2 \cos 60^\circ - F_3 \\ &= 70 \sin 10^\circ + 50 \cos 60^\circ - 30 \\ &= 7.1554 \text{ N} \end{aligned}$$

$$\begin{aligned} F &= \sqrt{F_x^2 + F_y^2} = \sqrt{(25.635)^2 + (7.1553)^2} \\ &= 27 \text{ N} \end{aligned}$$

$$\tan \theta = \frac{F_y}{F_x} = \frac{7.1553}{25.635} \Rightarrow \theta = 16^\circ$$



$\therefore$ the resultant force is 27 N, acting at an angle of 16° above the positive x-axis.

Note:

Answers should be left to the least s.f. (or one more) of the given data.

For example, in this question, the data given are all 2 s.f. values, hence the final answer(s) should be given in either 2 s.f. or 3 s.f..

However, this question specified that answers are to be given to 2 s.f..

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- C1 (a) (i)** It is NOT feasible to adopt system because it is not possible to express time in terms of the base quantities of mass, area and density.

- (ii)** Treating the Raffles System as SI base units:

Since work done $W = F \times d$, the metre, $m = \frac{\text{unit of } W}{\text{unit of } F} = \text{J N}^{-1}$

Since power $P = \frac{\text{energy}}{\text{time}}$, the second, $s = \frac{\text{unit of energy}}{\text{unit of } P} = \text{J W}^{-1}$

Since force $F = m a$, the kilogram,

$$\text{kg} = \frac{\text{unit of } F}{\text{unit of } a} = \frac{\text{N}}{\text{m s}^{-2}} = \frac{\text{N}}{\text{J N}^{-1}} \times (\text{J W}^{-1})^2 = \text{N}^2 \text{ J W}^{-2}$$

- (b) (i)** No. There are replicas of the International Prototype all over the world. Choosing one of these replicas to define the kilogram will not significantly alter the value of the kilogram.

- (ii)** Advantages:

1. The unified atomic mass constant is a physical constant. Adopting it as a base unit will allow the kilogram to be reconstructed anywhere in the world.
2. For the kilogram whose definition is based on a prototype, its definition will change when the prototype is replaced. This will not happen for the unified atomic mass unit, i.e. its definition is not as susceptible to change the way the kilogram is.

Disadvantage: It is too small a unit for daily life.